

Gruhesh Sri Sai Karthik Kurra

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ACADEMIC QUALIFICATION

Bachelor of Technology in Computer Science and Engineering

KL University, Hyderabad, India

August 2021 – May 2025

CGPA: 9.72/10

WORK EXPERIENCE

AI Integration Engineer

October 2025 – Present

InfoAjax Consulting Pvt Ltd, Remote, Hyderabad, India

- Leading development of AI-driven integration framework connecting enterprise applications with cybersecurity services through standardized APIs and intelligent orchestration
- Architected agent-based system using Azure OpenAI for automated schema mapping and adaptive connection strategies with performance learning from historical patterns
- Built full-stack application using React 18, Vite, TailwindCSS frontend and FastAPI backend with WebSocket integration for real-time monitoring and agent coordination
- Integrated Azure Active Directory via Microsoft Graph API for application discovery and automated metadata management

WSO2 API Developer

October 2024 – June 2025

InfoAjax Consulting Pvt Ltd, Remote, Hyderabad, India

- Developed and deployed RESTful APIs for PLDT (Philippines' largest telecommunications provider) using WSO2 Integration Studio for ticketing system integration (create fault ticket, update fault ticket, ticket management)
- Managed production assistance ticket systems with secure credential management through GitLab code repositories and Choreo cloud hosting platform with vault-based secrets management
- Performed systematic debugging in environments with rare WSO2 framework documentation, employing trial-and-error processes to resolve production issues; directly engaged with international clients under strict SLAs to deliver solutions

Founder & CEO

March 2024 – January 2025

Zynthetix (Personal Research Venture - Synthetic Data Generation Platform), Hyderabad, India

- Founded personal research project addressing dataset scarcity in ML training through synthetic data generation platform producing tabular, image, and text data using GANs and VAEs
- Designed innovative meta-controller architecture concept where a mother model orchestrates 200-1000 specialized child models (each expert in specific data types) to generate millions of rows without duplication at scale
- Implemented initial prototype with transformer-based PII detection (98.5% accuracy) and achieved 95% statistical similarity between synthetic and real data across correlation structures and distribution shapes
- Collaborated with 3 developers during prototyping phases for frontend development and data pipeline testing; established Git workflows and CI/CD pipelines (development paused due to high-end GPU infrastructure constraints required for training 200+ child model ensemble)

INTERNSHIP & TRAINING

Research Intern

December 2024 – June 2025

International Institute of Information Technology Hyderabad (IIITH), Hyderabad, India

- Received structured training in LLM optimization techniques including quantization, pruning, and knowledge distillation for efficient deployment on resource-constrained devices
- Led development of Layout Preserving Document Translation system: implemented Vision Transformer-based layout detection (92% accuracy identifying text regions, tables, images), designed character metrics-based font detection with K-means clustering for color palette extraction (88% accuracy)
- Built FastAPI backend with RESTful endpoints integrating Tesseract OCR for multi-language text extraction (English, Hindi, Telugu, German) while preserving document formatting, fonts, and colors
- Presented project at IIIT Hyderabad research expo (May 2025), conducting live demonstrations to faculty and industry visitors, receiving immediate feedback, and implementing real-time system improvements through iterative debugging

AI & ML Research Training

May 2024 – October 2024

iHub-Data, IIIT Hyderabad, Hyderabad, India

- Completed intensive 6-month apprenticeship program focused on AI and ML research methodologies, including hands-on training in neural architecture design, model optimization, and research best practices

- Engaged in hybrid on-site and remote training covering LLM fundamentals, prompt engineering, model fine-tuning, quantization techniques, and deployment strategies for production systems
- Worked on research projects involving document analysis, OCR systems, and computer vision applications under mentorship of IIIT Hyderabad faculty

Intern

July 2023 – October 2023

Xtraleap India Pvt Ltd, Remote, Hyderabad, India

- Worked on data science and analytics projects involving exploratory data analysis, statistical modeling, and machine learning model development using Python (pandas, NumPy, scikit-learn, Matplotlib)
- Gained practical experience in data preprocessing, feature engineering, and model evaluation on real-world datasets across retail analytics and customer segmentation domains

PROJECTS

BIE: Bit-Index Encoding for Neural Network Weight Compression

Duration: December 2024 – January 2025 **Team Size:** Individual

GitHub: github.com/GruheshKurra/bit-index-encoding-research

Summary: Novel weight compression framework encoding only positions of "1" bits in sparse weight matrices to achieve high compression ratios while maintaining model accuracy.

Individual Contribution:

- Implemented three encoding variants (binary for maximum compression, bitplane for progressive reconstruction, blocked for cache locality) in Python with custom sparse matrix kernels optimized using Numba JIT compilation
- Benchmarked across 253 experimental configurations on CNNs and transformer architectures, achieving 40× compression at 95% sparsity with MSE < 10⁻⁶ while supporting direct computation without decompression
- Outperformed traditional pruning and quantization baselines on both compression ratio and inference speed metrics; published research with complete open-source implementation

Hybrid RAG-Enhanced Deepfake Detection

Duration: July 2024 – August 2024 **Team Size:** Individual

Website: zenodo.org/records/16732053

Summary: Privacy-first deepfake detection system combining Retrieval-Augmented Generation with visual anomaly detection for improved reliability and interpretability.

Individual Contribution:

- Designed two-stage architecture: RAG pipeline using FAISS vector database with CLIP-based visual encoding for semantic retrieval of known deepfake patterns, combined with custom CNN for visual inconsistency detection (facial artifacts, asymmetry, temporal incoherence)
- Implemented uncertainty-aware classification using Monte Carlo Dropout providing confidence scores alongside predictions; achieved 94.8% accuracy and 94.6% F1-score through local processing prioritizing privacy

Layout Preserving Document Translation

Duration: December 2024 – June 2025 **Team Size:** 2

Video: youtu.be/rcvSuBcBjyg

Summary: Comprehensive PDF document analysis and translation system preserving original formatting, fonts, colors, and layout using Vision Transformers and OCR technologies.

Individual Contribution:

- Implemented ViT-based layout detection achieving 92% accuracy in identifying text regions, tables, and images; developed character metrics-based font detection with K-means clustering for color extraction (88% accuracy)
- Built FastAPI backend with RESTful endpoints integrating Tesseract OCR for multi-language support (English, Hindi, Telugu, German); conducted live demos at IIIT Hyderabad research expo with real-time user feedback integration

Clinical Trial Similarity Analysis

Duration: September 2024 **Team Size:** Individual

GitHub: github.com/GruheshKurra/Clinical-Trial-Similarity-Analysis

Summary: Semantic grouping and retrieval system for clinical studies using PubMedBERT and advanced NLP techniques.

Individual Contribution:

- Implemented PubMedBERT-based system for domain-specific medical embeddings achieving 99.5% similarity accuracy and 0.891 top-10 retrieval precision
- Developed GPU-accelerated processing pipeline optimized for NVIDIA RTX 3090 with efficient batch processing for large-scale clinical trial databases

GPT-2 Implementation from Scratch

Team Size: Individual

GitHub: github.com/GruheshKurra/FirstGPTFromScratch

Summary: Complete implementation of GPT-2 architecture (124M parameters) with custom tokenization, multi-

head attention, and transformer blocks; pretrained from scratch demonstrating understanding of large language model fundamentals.

Transformers from Scratch

Team Size: Individual

GitHub: github.com/GruheshKurra/TransformersFromScratch

Summary: Complete implementation of Transformer architecture from "Attention Is All You Need" paper with multi-head attention and positional encoding; achieved 85%+ accuracy on sentiment classification (200K parameters, 10-minute training on GPU).

Graph Neural Networks Suite

Team Size: Individual

GitHub: github.com/GruheshKurra/GraphNeuralNetworks-GNN-

Summary: Implemented and compared GCN (79.3%), GraphSAGE (76.8%), and GAT (81.9%) architectures on Cora citation dataset for node classification using PyTorch Geometric.

Diffusion Models from Scratch

Team Size: Individual

GitHub: github.com/GruheshKurra/DiffusionModelFromScratch

Summary: Complete DDPM architecture with forward/reverse diffusion processes; achieved efficient 30-min GPU training with 130K parameters for MNIST generation.

Generative Adversarial Networks (GANs)

Team Size: Individual

GitHub: github.com/GruheshKurra/GAN_Implementation

Summary: Complete GAN architecture for MNIST digit generation with cross-platform GPU support (MPS, CUDA, CPU); 576K-3.5M parameters with multiple training modes.

Variational Autoencoders (VAE)

Team Size: Individual

GitHub: github.com/GruheshKurra/VariationalAutoencoders

Summary: Complete VAE with encoder, decoder, and reparameterization trick; achieved clear digit reconstructions with organized latent space (400K parameters).

MySQL Query Generator (NL2SQL)

Team Size: Individual

GitHub: github.com/GruheshKurra/nl2sql-pretrained

Summary: GPT-style transformer for natural language to SQL conversion; lightweight model with 29.8M parameters achieving 1.42 perplexity in 12-minute training.

FARMCare AI Agritech Platform

Team Size: Individual

GitHub: github.com/GruheshKurra/FARMCare

Summary: Comprehensive agricultural platform using React and Google Gemini 1.5 Flash API with AI-powered plant health analysis, pest identification, soil analysis, and multilingual support (English, Hindi, Telugu).

AI Dashboard Creator

Duration: January 2025 – Present

Video: youtu.be/Nh_EkDAuPqs

Summary: AI-driven platform that automatically creates insightful dashboards from CSV files, with data privacy protection (98.5% PII detection accuracy) and intelligent image organization capabilities (95% classification accuracy across 50+ categories).

Visionary AI Platform

GitHub: github.com/GruheshKurra/CapstoneProject1 **Website:** visionaryapp.vercel.app

Video: youtu.be/HAOBZ3bEjMg

Summary: Comprehensive AI platform offering ChatWithPDF, DeepFake Detection, Image Generation, Object Detection, and Website Generator services; built with Vite.js, React Native, and Supabase.

E-Commerce Website (JSP & Java EE)

Video: youtu.be/aX84r7oQyto

Summary: Full-featured eCommerce web application with user authentication, shopping cart, admin dashboard, real-time chat support, and notification system using JSP, Java EE, and MySQL.

Blog App with FARM Stack

Video: youtu.be/KgUnAuoVnMQ

Summary: Full-stack blogging platform with secure JWT authentication, rich text editor, comment system, and responsive design using FastAPI, React, MongoDB, and TailwindCSS.

FastAPI Chat Application

GitHub: github.com/GruheshKurra/FastAPI_ChatApp

Video: youtu.be/g6c9G-eQ9JY

Summary: Real-time chat application using FastAPI and WebSocket with MongoDB for efficient message storage

and secure user authentication.

Additional ML/DL Implementation Projects

- **Anomaly Detection Suite** ([GitHub](#)) – Implemented 5 algorithms (Z-score, Isolation Forest, One-Class SVM, LOF, Autoencoder) with comprehensive evaluation
- **Clustering Algorithms** ([GitHub](#)) – K-Means, Hierarchical, DBSCAN, GMM for customer segmentation
- **Dimensionality Reduction Suite** ([GitHub](#)) – PCA, t-SNE, UMAP, Autoencoders comparison
- **Attention Mechanisms** ([GitHub](#)) – Multi-head attention implementation achieving 96%+ accuracy on Iris dataset
- **Sequence Modeling Suite** ([GitHub](#)) – RNN, LSTM, GRU, and Transformer implementations
- **Gesture-Based Mouse Control** ([GitHub](#)) – Hand gesture recognition using MediaPipe and OpenCV

PUBLICATIONS

- **Gruhesh Sri Sai Karthik Kurra**, Chandanasree Moparthi, Prathipati Manish Chowdary, and Pavan Kumar Pagadala, "Morphology-Aware Nested Character Embeddings for Word Representation," *2025 IEEE 7th International Conference on Cybernetics, Cognition and Machine Learning Applications (ICCCMLA)*, Accepted for publication, Forthcoming 2025
- **Gruhesh Sri Sai Karthik Kurra**, Koduru Sushma Reddy, Kasula Ranga Tanuj, and Koduru Srujana Reddy, "Dynamic Auto-Finetuning of Language Models Based on Confidence-Driven Knowledge Integration," *International Journal of Novel Research and Development (IJNRD)*, vol. 9, no. 11, pp. b905-c905, Nov. 2024, ISSN: 2456-4184
- **Kurra Gruhesh Sri Sai Karthik Kurra**, Kasula Ranga Tanuj, Yashwanth Adepu, Chandanasree Moparthi, and Nikhil Choudary Mukkamala, "Global Remote RAM Sharing: A Novel Framework for Distributed Computational Systems," *International Journal of Novel Research and Development (IJNRD)*, vol. 9, no. 12, pp. c635-c635, Dec. 2024, ISSN: 2456-4184
- **Kurra Gruhesh Sri Sai Karthik Kurra**, Koduru Sushma Reddy, Veluvolu Sai Venkata Deepak, Prathipati Manish Chowdary, and Nikhil Choudary Mukkamala, "Voice-Activated AI for Seamless Computer Interaction," *International Journal of Novel Research and Development (IJNRD)*, vol. 10, no. 1, pp. a698-c698, Jan. 2025, ISSN: 2456-4184
- **Gruhesh Sri Sai Karthik Kurra**, "BIE: Bit-Index Encoding for Efficient Neural Network Weight Compression," *Zenodo*, Sep. 2025, DOI: 10.5281/zenodo.17217218
- **Gruhesh Sri Sai Karthik Kurra**, "Hybrid RAG-Enhanced Deepfake Detection: A Novel Approach Combining Retrieval-Augmented Generation with Visual Inconsistency Analysis," *Zenodo Preprint*, Aug. 2024, DOI: 10.5281/zenodo.16732053

AWARDS & ACHIEVEMENTS

- **1st Place – University Webathon**, KL University, Hyderabad, India; 2022
Developed comprehensive diet management platform with nutrition tracking, meal planning, and health analytics in 4-hour competitive hackathon
- **2nd Place – Design Expo**, KL University, Hyderabad, India; 2022-2023
Created low-cost Arduino-based smart switchboard for IoT home automation with mobile app control and real-time monitoring as part of Design Thinking for Innovation Course
- **National Tennis Ranking of 395 in India**, All India Tennis Association (AITA); 2014-2019
Demonstrated athletic excellence, competitive discipline, and effective time management while maintaining academic performance
- **Technical Lead**, Student Activity Council, KL University, Hyderabad, India; 2023-2025
Mentored 20+ juniors in web development (React.js, Node.js, FastAPI), machine learning fundamentals, and project management; organized technical workshops and coding bootcamps; awarded SAC Momentum Award for technical leadership and community building contributions

PRESENTATIONS

- **"Layout Preserving Document Translation"**, IIIT Hyderabad Research Expo, Hyderabad, India; May 2025
Conducted live demonstrations to research faculty, PhD students, and industry visitors; performed real-time system testing with user-provided PDFs in multiple languages (Telugu, Hindi, German, English); incorporated immediate feedback through iterative debugging and system improvements on the spot

CERTIFICATIONS

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|---|---|
| • Google TensorFlow Developer Certificate
Google, Coursera, 2024 | • Amazon Web Services, Online, 2024 |
| • AWS Certified Cloud Practitioner
Amazon Web Services, Online, 2024 | • Automation Anywhere Certified Advanced Automation Professional
Automation Anywhere, Online, 2024 |
| • AWS Certified Solutions Architect Associate | • Oracle Cloud Infrastructure Architect Associate |

- Oracle University, Online, 2024
- Oracle Cloud Infrastructure Generative AI Professional Oracle University, Online, 2024
- RedHat Certified Enterprise Application Developer RedHat, Online, 2024
- WSO2 Certified Micro Integrator Practitioner V4 WSO2, Online, 2024
- WSO2 Certified Micro Integrator Developer V4 WSO2, Online, 2024

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, Java, C, C++, R, HTML, CSS, PHP, SQL, MySQL

AI/ML Frameworks & Tools: TensorFlow, PyTorch, PyTorch Geometric, Hugging Face Transformers, LangChain, LlamaIndex, scikit-learn, Gradio, Streamlit, UMAP, Azure OpenAI API, OpenAI GPT Models, Ngrok

Web & Mobile Development: React.js (React 18), Next.js, Vue.js, Vite, FastAPI, Node.js, Express.js, React Native, Expo, Kivy, TailwindCSS, ShadcnUI, AceternityUI, Material-UI, Uvicorn

Cloud & DevOps: AWS (EC2, S3, Lambda, Solutions Architect), Azure (Active Directory, OpenAI Service, Microsoft Graph API), Google Cloud Platform, Vercel, Git, GitLab, WSO2 Integration Studio, Docker, CI/CD Pipelines, RESTful API Development, Serverless Architecture, Choreo

Databases & Authentication: PostgreSQL, MongoDB, MySQL, SQLite, Supabase, Firebase, Redis, JWT, Auth0, Clerk, Azure Active Directory

Real-Time & Communication: WebSocket, Pydantic, AsyncIO, Threading, gRPC, Real-time Data Streaming

AI/ML Specializations:

- Large Language Models:** GPT-2 from scratch (124M parameters), LLM Optimization, Fine-tuning, Prompt Engineering, Token Classification
- Generative Models:** GANs (576K-3.5M parameters), VAEs (400K parameters), Diffusion Models (DDPM, 130K parameters), Image Generation
- Transformer Architecture:** Multi-head Attention, Positional Encoding, Sequence Modeling, Sentiment Analysis (85%+ accuracy)
- Graph Neural Networks:** GCN (79.3%), GraphSAGE (76.8%), GAT (81.9%), Node Classification
- Model Compression:** Quantization, Pruning, Knowledge Distillation, Bit-Index Encoding (40× compression)
- Computer Vision:** OCR (Tesseract), Vision Transformers (ViT), Object Detection, Image Classification, Deep-Fake Detection (94.8% accuracy)
- Natural Language Processing:** NL2SQL (29.8M parameters, 1.42 perplexity), Text Classification, Sentiment Analysis, PubMedBERT
- Advanced Techniques:** Retrieval-Augmented Generation (RAG), Uncertainty Quantification (Monte Carlo Dropout), Synthetic Data Generation
- Machine Learning:** Anomaly Detection (5 algorithms), Clustering (K-Means, DBSCAN, GMM), Dimensionality Reduction (PCA, t-SNE, UMAP)
- Sequence Models:** RNN, LSTM, GRU implementations for time series and sequential data

Data Science & Analysis: Pandas, NumPy, Matplotlib, Seaborn, Tableau, Dataset Preprocessing, Data Privacy (98.5% PII detection), Data Visualization, Statistical Analysis, Synthetic Data Generation, Performance Metrics, Model Comparison, Elbow Method, Silhouette Analysis

MLOps & Deployment: Model Deployment, Model Optimization, Model Serving, AI Model Hosting, MLflow, Model Checkpointing, Comprehensive Logging, Cross-Platform Optimization (MPS, CUDA, CPU)

Cybersecurity & Integration: AI-Driven Security Orchestration, Policy-Based Automation, Identity & Access Management (IAM), Security Information and Event Management (SIEM), Azure AD Integration, Audit Logging, Compliance Monitoring, Agent-Based Architecture

System Design & Architecture: Microservices Architecture, Distributed Systems, Agent Pool Management, Orchestration Patterns, API Gateway Design, Event-Driven Architecture, Scalable System Design

EXTRACURRICULAR ACTIVITIES

- Volunteer, StreetCause (Social Service NGO),** Hyderabad, India; 2023 – 2024
Participated in community outreach programs supporting underprivileged communities through educational initiatives and resource distribution; contributed to organizing awareness campaigns on digital literacy and technology access; demonstrated commitment to social responsibility and community development; Certificate of Service awarded
- Technical Lead, Student Activity Council (SAC),** KL University, Hyderabad, India; 2023 – 2025
Mentored 20+ juniors in web development and machine learning fundamentals; organized technical workshops, coding bootcamps, and hackathons for 100+ students; developed SAC website with database for user management and automated certificate generation; managed team of 5 coordinators; awarded SAC Momentum Award for technical leadership and community building contributions